



# Lesson 5: Experimental Methodology

Technologies for Artificial Intelligence

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# Before we start

- Exercise 4, which we discuss now
- The threshold you chose, and how you chose it

# We broke a promise on purpose

- Lesson 4 chose a threshold **by looking at the test set**
- Lesson 1 forbade exactly that
- Today we pay the debt

# Today: how to tell whether it works

- One split is a lottery, and how wide the lottery is
- Cross-validation: what it estimates, and what it does not
- Bias, variance, and the floor you cannot go below
- Learning curves: which fix to buy
- Leakage that survives cross-validation

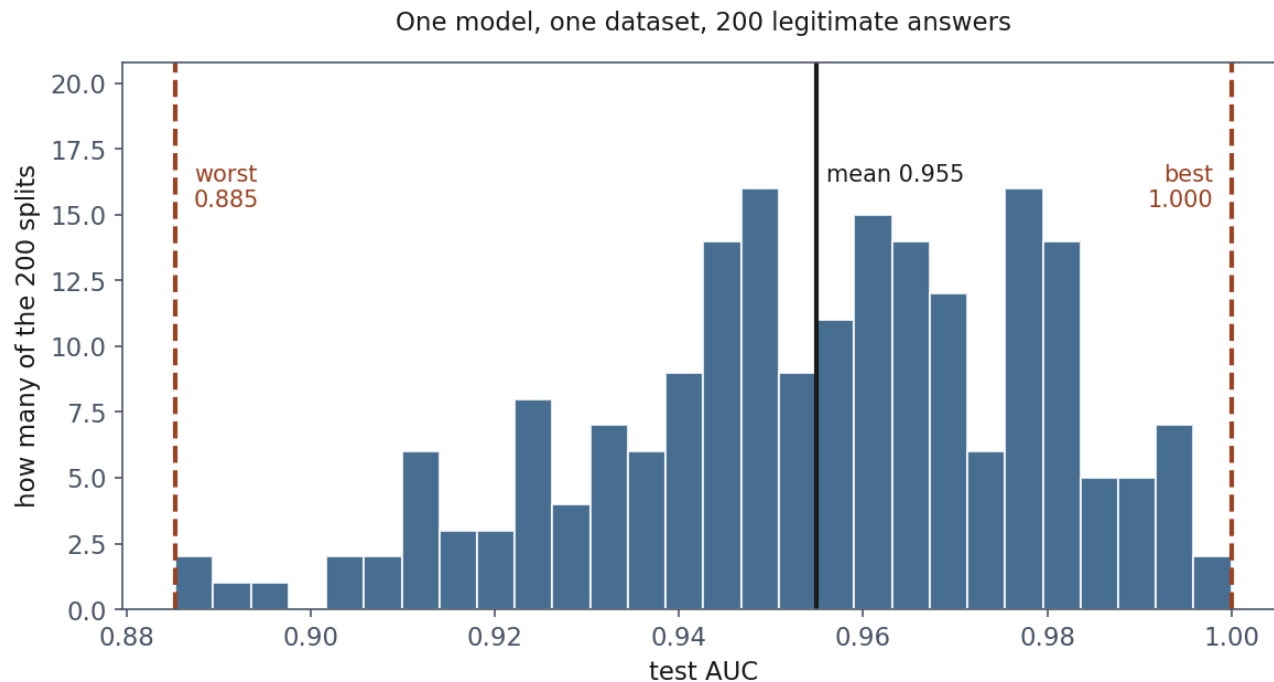
# A test score is a measurement

- You would not report a poll of forty people to three decimals
- Holding data out makes the estimate **unbiased**
- It does nothing to make it **precise**

# The fleet, cut down

- Lesson 4's drives, reduced to **800**, of which **29 fail**
- Not to exaggerate: 800 rows with 29 positives is the ordinary case

# 200 honest splits, one dataset



# Why so wide?

- Each test set holds **7 failures**
- Every AUC (area under the receiver operating characteristic curve) rests on seven

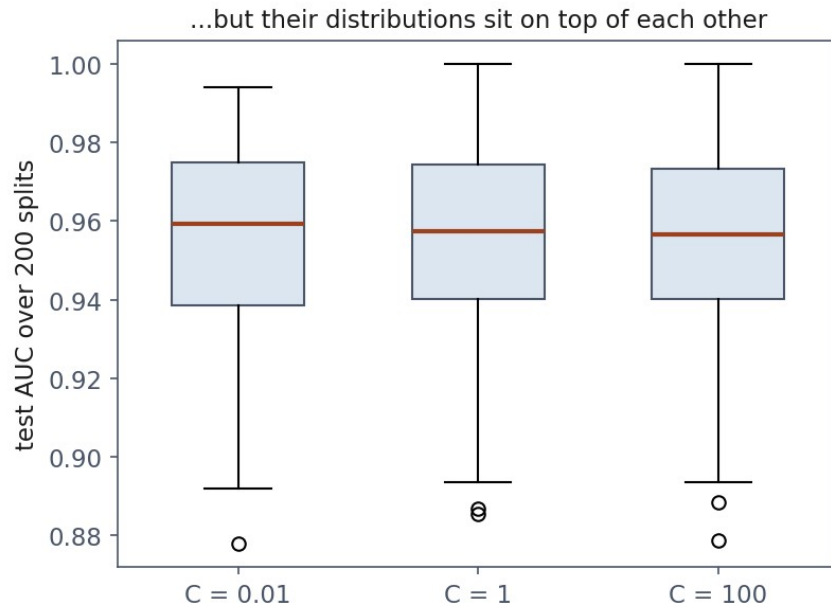
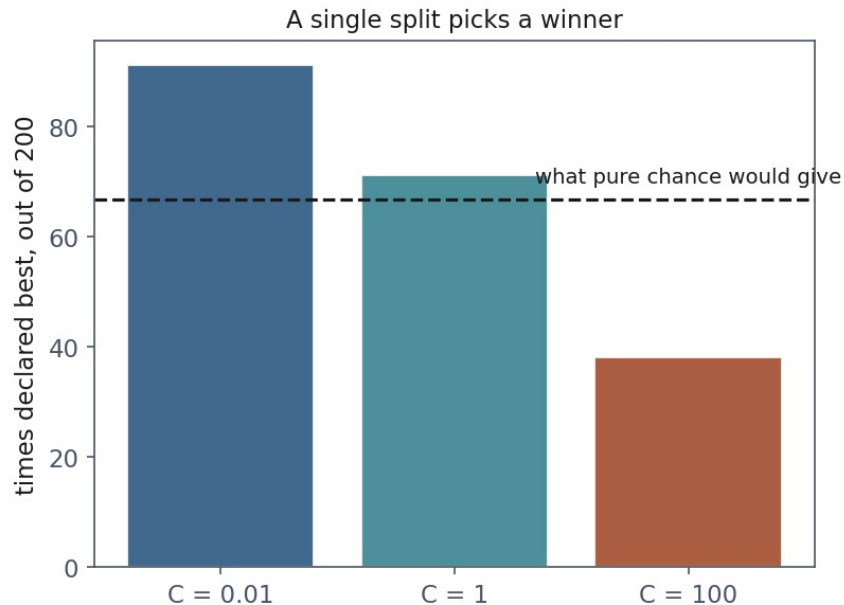
# Parameters and hyperparameters

- **Parameters** — estimated by the fit:  $w$ ,  $b$
- **Hyperparameters** — not estimated: penalty  $\lambda$ , neighbours, depth, threshold
- Not “set before the fit”: the threshold is chosen after it
- Training error would prefer  $\lambda = 0$ , so they are chosen by **comparison**, on unseen data
- scikit-learn’s  $C$  is a **reciprocal** penalty:  $C = 0.01$  is the strongest

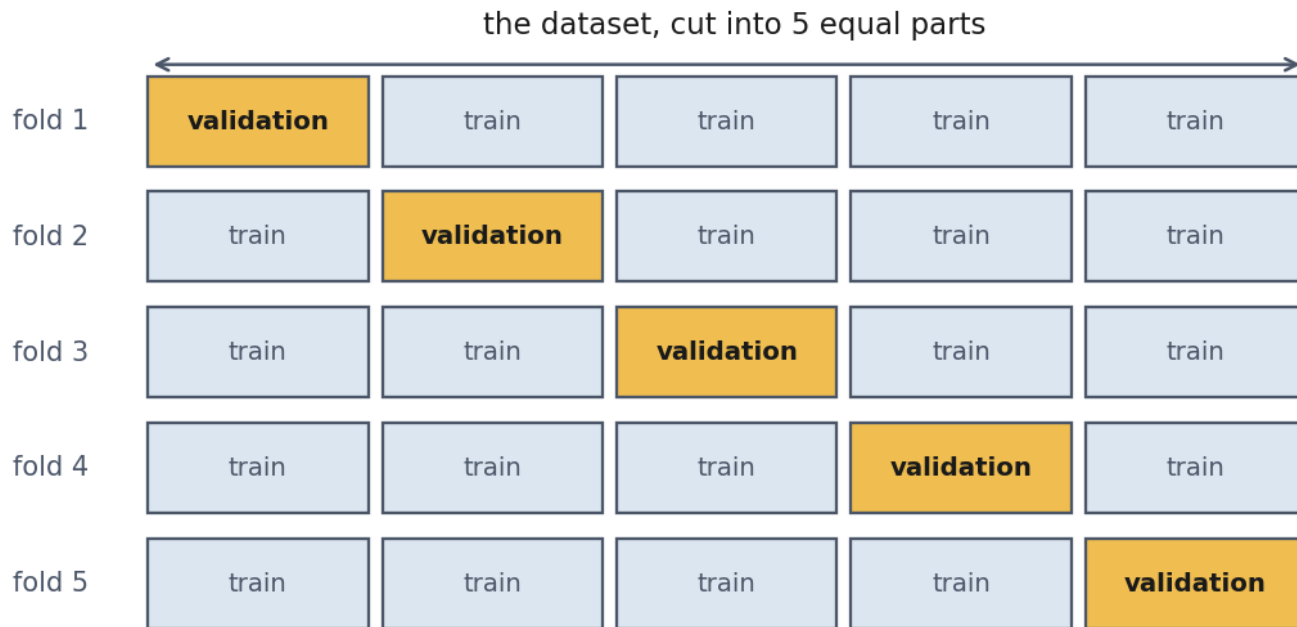
# Now the expensive part: choosing

| Model      | Mean AUC | Times declared best |
|------------|----------|---------------------|
| $C = 0.01$ | 0.9549   | 91                  |
| $C = 1$    | 0.9549   | 71                  |
| $C = 100$  | 0.9544   | 38                  |

# The same three models, drawn honestly



# k-fold cross-validation



# What it actually estimates

- The performance of a **procedure**, not of one fitted model
- Slightly **pessimistic**: each fold trains on 80% of the data

# Report the spread, but not as a confidence interval

- Any two training folds share **three quarters** of their rows
- The scores are correlated, so the usual standard error is optimistic

# Stratify when a class is rare

| Fold | Plain KFold             | StratifiedKFold |
|------|-------------------------|-----------------|
| 1    | 7 failures of 160       | 5 of 160        |
| 2    | <b>1 failure of 160</b> | 6 of 160        |
| 3    | 7 of 160                | 6 of 160        |
| 4    | 5 of 160                | 6 of 160        |
| 5    | 9 of 160                | 6 of 160        |

# How much does it help?

|        | Single split | 5-fold CV |
|--------|--------------|-----------|
| worst  | 0.9119       | 0.9439    |
| best   | 1.0000       | 0.9588    |
| mean   | 0.9600       | 0.9532    |
| spread | 0.0881       | 0.0149    |

# How many folds?

- **k = 5 or 10** in practice
- Larger k: less pessimistic, more expensive, folds more correlated
- k = m is **leave-one-out**

# When the rows are not independent

- Ten readings from the **same drive**, split at random
- Nine in training, one in test
- The model recognises the drive, not the failure

# Split by the thing you must generalise to

- Same entity in two folds → **GroupKFold**
- Predicting the future → **TimeSeriesSplit**

# Notebook 1, live

- The split lottery, the three identical models, k-fold by hand

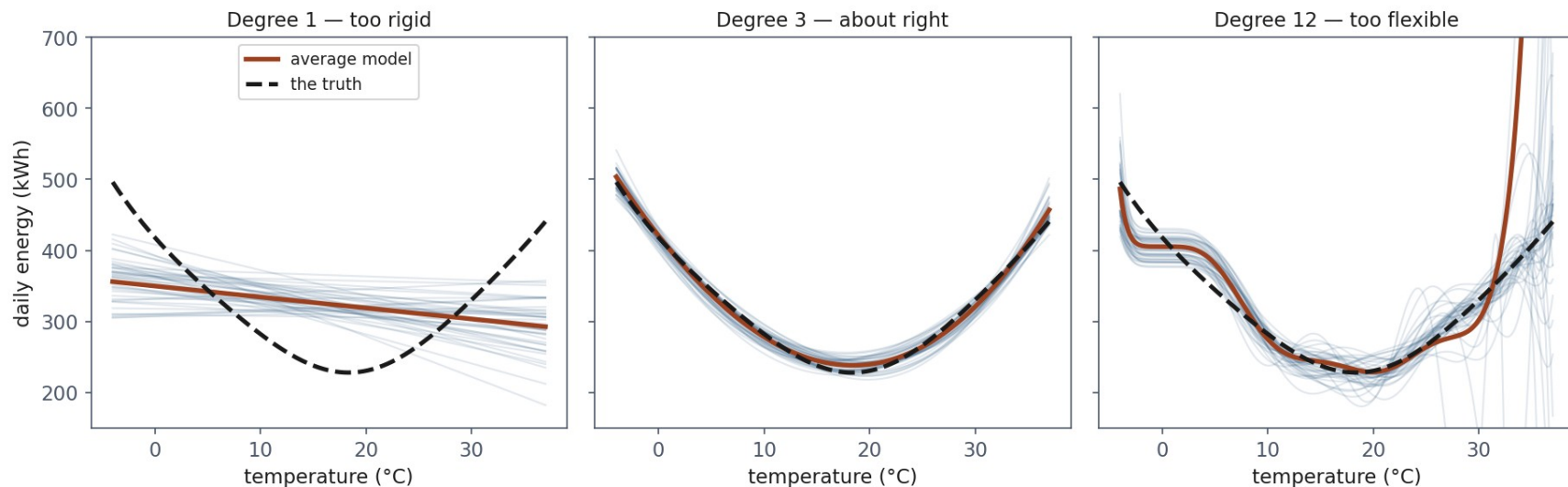
# Break

- Ten minutes

# Where does the error come from?

- **Bias**: the average model is wrong
- **Variance**: the individual models disagree
- **Noise**: the target itself is uncertain

# Three hundred parallel universes



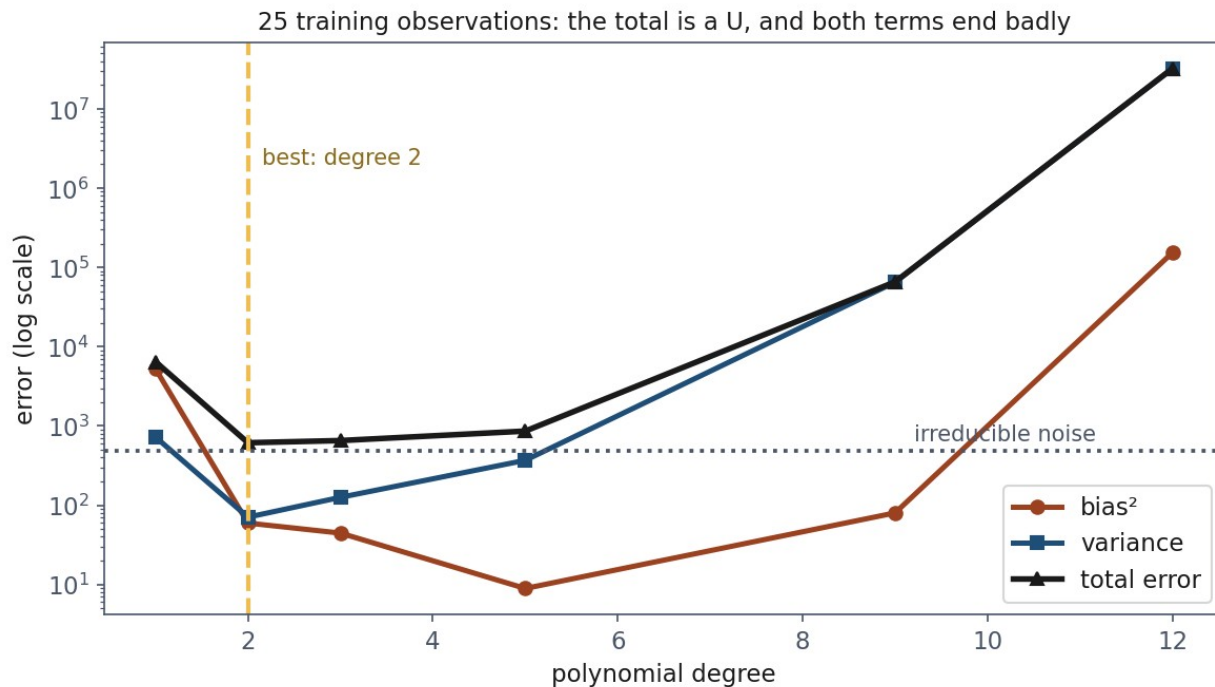
# The decomposition

$$E[(y - \hat{f})^2] = \text{bias}^2 + \text{variance} + \sigma^2$$

# Measured, on 25 observations

| Degree | bias <sup>2</sup> | Variance     | Noise | Total        |
|--------|-------------------|--------------|-------|--------------|
| 1      | 5,250.8           | 723.8        | 484.0 | 6,458.6      |
| 2      | 59.6              | 70.8         | 484.0 | <b>614.3</b> |
| 5      | 8.9               | 368.9        | 484.0 | 861.8        |
| 12     | 153,537.3         | 32,174,344.9 | 484.0 | 32,328,366.2 |

# The picture everyone has seen, now measured



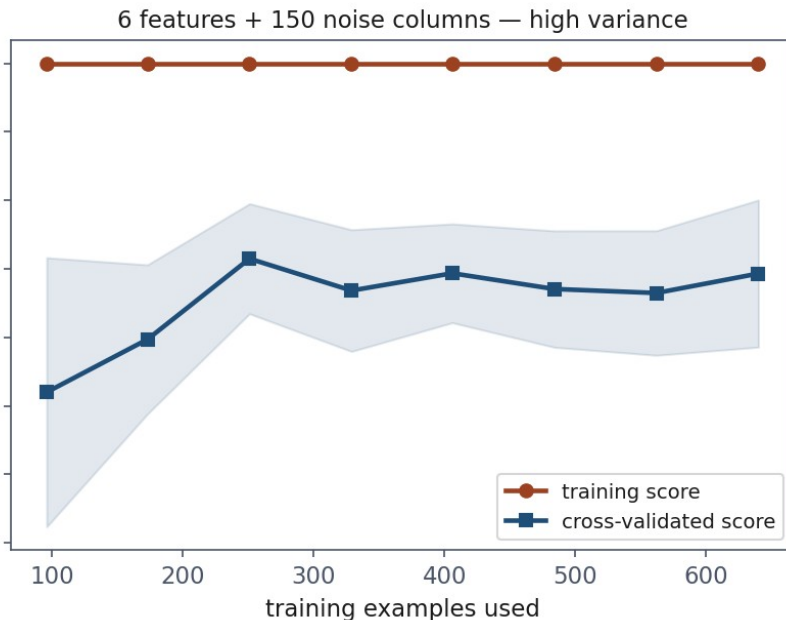
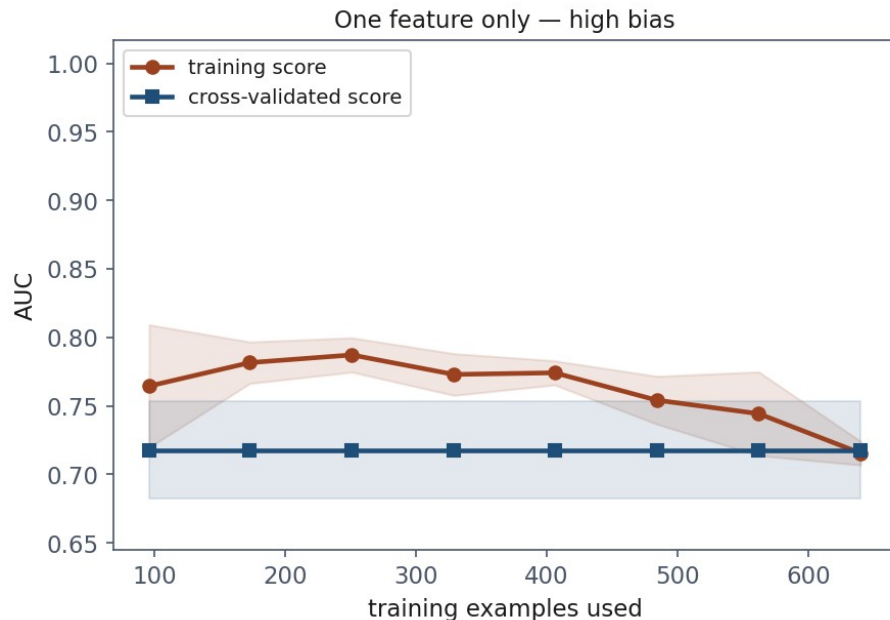
# The floor you cannot go below

- Noise variance is **484** at every complexity
- A reported error below the floor means **contamination**, not excellence

# The best model depends on how much data you have

| Degree | m = 25       | m = 60       | m = 150      |
|--------|--------------|--------------|--------------|
| 2      | <b>614.3</b> | 570.6        | 555.3        |
| 5      | 861.8        | <b>540.9</b> | <b>506.9</b> |
| 12     | 32,328,366.2 | 7,626.9      | 899.0        |

# Learning curves: the same diagnosis, without knowing the truth



# Which fix to buy

- Curves **meet low** → bias. More data will **not** help
- Wide gap, validation **still rising** → more data **will**
- Wide gap, validation **flat** → regularise, drop features. **Ours is this one**

# Notebook 2, live

- Measuring bias and variance; the learning curves

# Now: what cross-validation does not protect you from

- 2,000 columns of **pure random noise**
- No relationship to the label whatsoever
- The honest score of any model on it is **0.500**

# The code almost everyone writes

```
sel = SelectKBest(f_classif, k=10)
sel.fit(noise, y)
X = noise.loc[:, sel.get_support()]
cross_val_score(model, X, y, cv=cv)
```

# 0.931

- Cross-validated AUC **0.931** on data with no signal
- Four of five folds between 0.93 and 0.98

# Fix it, and look what happens

- Selection moved **inside** the pipeline, refitted per fold
- 20 different fold seeds: **0.658**
- The truth is still 0.500

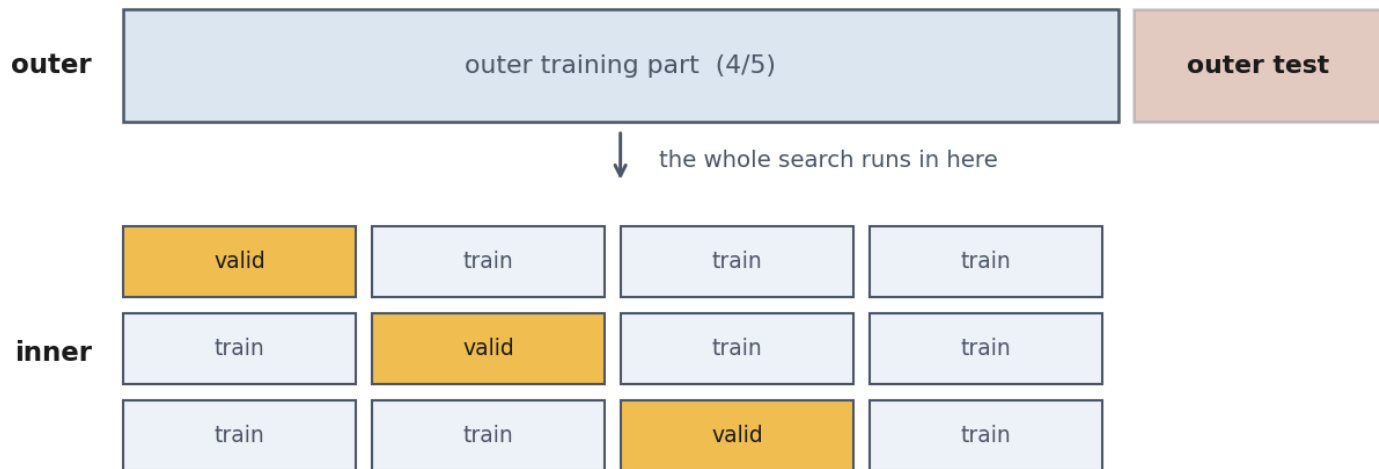
# Why it does not return to 0.5

- With 2,000 columns and 800 rows, some columns match the label **by accident**
- That accident belongs to the **sample**, not to the split
- Every fold contains it

# Hyperparameter search is the same problem

|                            | AUC           |
|----------------------------|---------------|
| best of 25 candidates      | <b>0.7999</b> |
| average candidate          | 0.7265        |
| nested cross-validation    | 0.6699        |
| the truth, by construction | 0.5           |

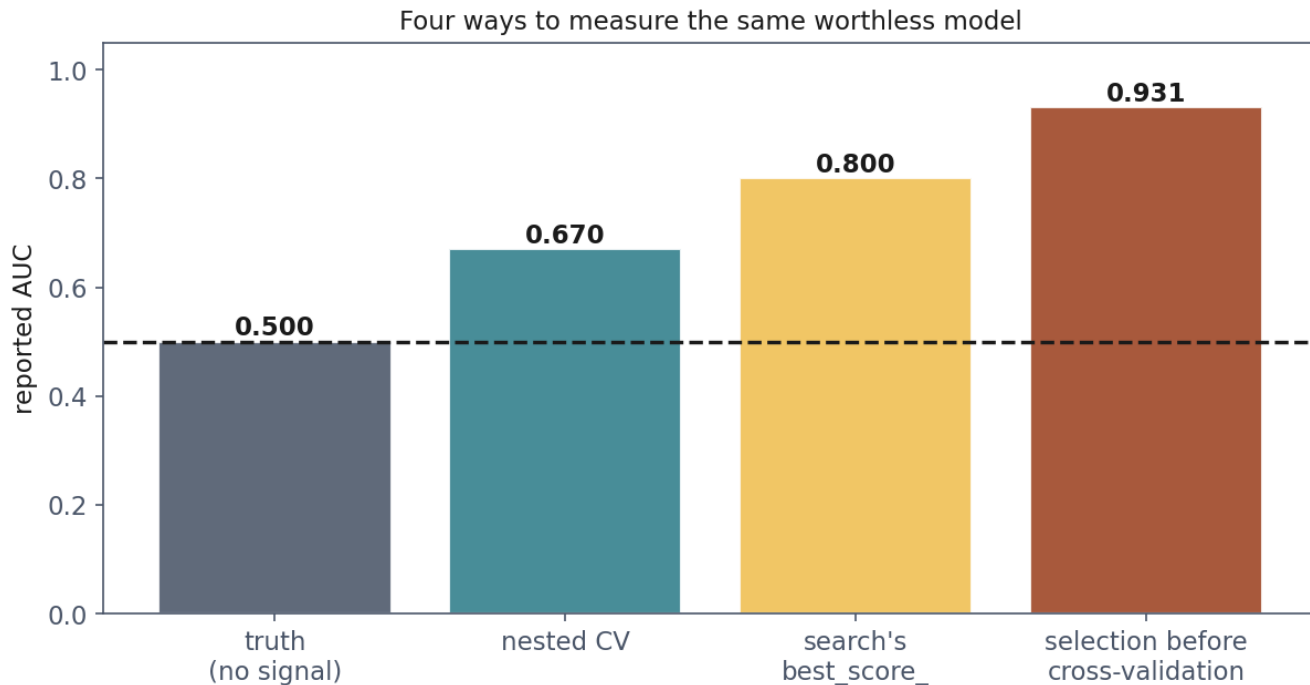
# Nested cross-validation



The inner loop may overfit itself as much as it likes.

**The outer test block was never part of it: that is the number you report.**

# Four ways to measure the same worthless model



# The debt, paid

| Policy                                | Cost on the test set |
|---------------------------------------|----------------------|
| threshold 0.50, the default           | 10,540 EUR           |
| threshold chosen on <b>validation</b> | <b>4,700 EUR</b>     |
| threshold chosen on <b>test</b>       | 3,300 EUR            |

# Three sets, three jobs

- **Train:** fit the parameters
- **Validation:** make every choice: model, hyperparameters, threshold
- **Test:** report. Touched **once**

# A fixed seed buys repeatability, not stability

- 30 seeds, all perfectly reproducible: **0.913 to 1.000**
- AUC 1.000 (`random_state=3`) is reproducible **and** misleading

# What to fix, and what to write down

- Every `random_state`, and  
`np.random.default_rng(seed)`
- **Library versions**: defaults change between releases
- What you could **not** fix: threads, GPU non-determinism

# Notebook 3, live

- The 0.931, the fix, the nested search

# What to put in a report

- The metric, the **spread**, and how it was estimated
- How many **positives** the test set held
- Every seed, and every library version
- Which choices were made on which data

# What to take away

- **A test score is a measurement.** 0.885 to a perfect **1.000**, same model
- **Choosing on one split chooses noise**
- **bias<sup>2</sup> + variance + noise** is an identity, and the noise floor never moves
- **Curves that meet low mean bias:** more data will not help
- **Cross-validation protects against fold leakage, nothing more**

# Homework

- **Exercise 5**, discussed at the start of **Friday 6 November**
- A result that is too good. Find out why, then produce the honest number